

# Day 7 – Webhook Fundamentals

## (Synchronous)

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### Setup & Prerequisites

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1. Setup Checklist:
- **Virtual Environment active:** Ensure your terminal shows `(.venv)` in the prompt.
  - **Django Server running:** Run the dev server to keep configurations loaded.
  - **Webhook Test URL:** Open a browser page to `https://webhook.site` to get a temporary test URL.
  - **IDE Ready:** Open `games/webhooks.py` and `games/management/commands/check_expired_keys.py` in your IDE.
2. Prerequisites:
- You must have completed Day 6 successfully (Management command structured and working locally to expire keys).

### Learning Objectives

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- By the end of this class, you will be able to:
- Explain the concept of Webhooks (HTTP callbacks) and contrast them with polling.
  - Implement cryptographic payload signing using HMAC-SHA256.
  - Enforce deterministic JSON serialization using `sort_keys=True` in `json.dumps()`.
  - Use the `requests` library to make synchronous POST requests with custom timeouts and headers.
  - Identify the core limitations of synchronous HTTP calls in management commands.

### Classroom Timeline (90 min)

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| Segment                      | Duration | Focus                                                                               |
|------------------------------|----------|-------------------------------------------------------------------------------------|
| 1. Hook & Big Picture        | 20 min   | Explain the webhooks model. Contrast polling vs. event-driven callbacks.            |
| 2. Key Concepts & Core Ideas | 20 min   | Explain payload verification, HMAC, constant-time comparison, and JSON key sorting. |

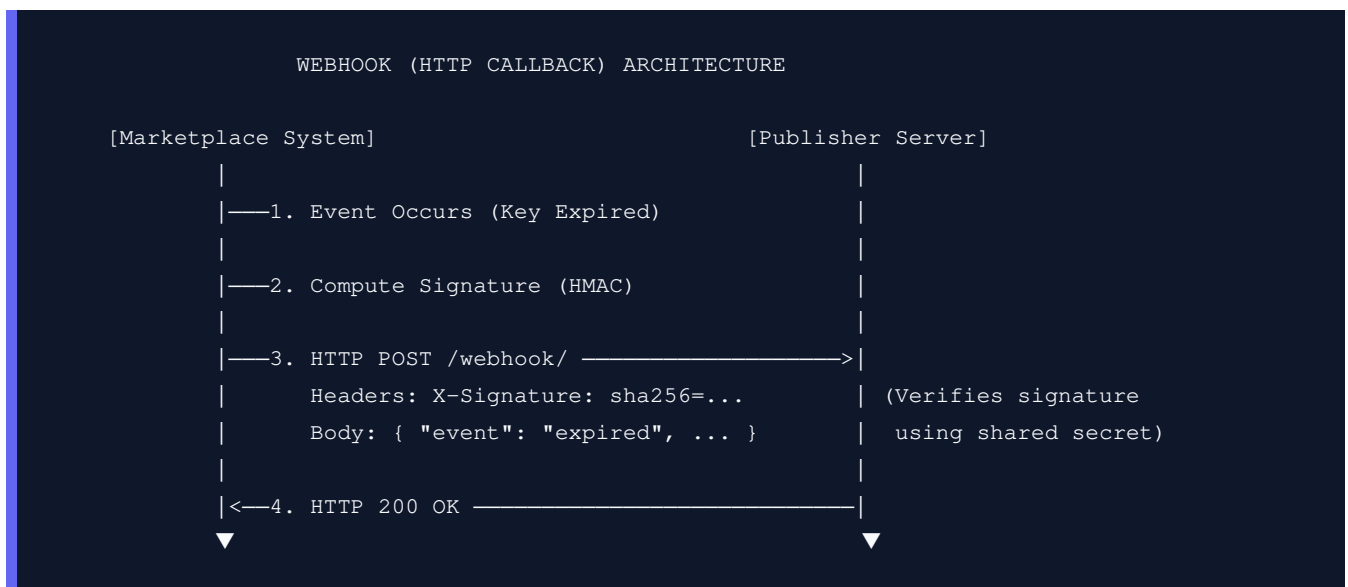
|                                        |        |                                                                                           |
|----------------------------------------|--------|-------------------------------------------------------------------------------------------|
| <b>3. Live Coding Walkthrough</b>      | 30 min | Write the synchronous webhook helper, integrate it with the management command, and test. |
| <b>4. Discussion &amp; Limitations</b> | 10 min | Analyze the failures of synchronous webhooks (blocking, timeout delays, no retry engine). |
| <b>5. Check for Understanding</b>      | 10 min | Q&A on security signatures, response codes, and sync blockages.                           |

## Lecture Step-by-Step Delivery Plan

### 1. Hook & Big Picture (20 min)

Polling vs Webhooks analogy: *"Imagine checking your physical mailbox every 5 minutes to see if a package arrived (polling) versus having a delivery person ring your doorbell when it arrives (webhook). Webhooks are push notifications for APIs."*

- **Security Risk:** Since any client can send a POST request to the publisher's public webhook URL, the publisher needs a way to guarantee the request actually came from our marketplace. We solve this using cryptographic payload signing.



### 2. Key Concepts & Core Ideas (20 min)

Introduce these webhook security and network concepts:

- **HMAC-SHA256:** Hash-based Message Authentication Code. Both systems share a secret key. We sign the payload bytes with this secret, producing a signature hex digest. The publisher recalculates the signature using the same secret and matches them.
  - **Question:** Why sign? Because anyone can POST to the publisher's webhook URL.

■ *Answer:* A signature verified with a shared secret ensures the payload is authentic, has not been tampered with in transit, and was not sent by an attacker.

- **sort\_keys=True in JSON:** JSON objects are unordered maps. If keys are serialized in different orders on the sender and receiver (e.g. {"key": 1, "game": 2} VS {"game": 2, "key": 1}), the raw byte sequences will differ, causing the HMAC signatures to mismatch. Sorting keys makes serialization deterministic.

○ **Question:** Why do we pass `sort_keys=True` to `json.dumps()`?

■ *Answer:* JSON key order is not guaranteed. Sorting makes the serialized string deterministic, ensuring signature calculations match.

- **hmac.compare\_digest():** A utility that compares digests in constant time. This prevents timing attacks (where attackers deduce correct signature bytes by measuring tiny variations in execution time).
- **Outbound Timeouts:** Why we must always set a timeout (e.g. `timeout=5`) when calling external services. Without a timeout, a hanging remote server will cause our command line process to block indefinitely.
  - **Question:** Why do we set a timeout on requests?
    - *Answer:* Without a timeout, a hanging remote server will cause our script to block indefinitely, consuming system resources and delaying other tasks.
- **raise\_for\_status():** A method that raises an HTTP error if the server returns a 4xx or 5xx error code.

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### 3. Live Coding Walkthrough (30 min)

#### Step 3.1: Implementing the Webhook Sender

- **Concept:** Create `games/webhooks.py`. We will write a helper function that constructs the expiry event payload, signs it using HMAC-SHA256, sets the custom headers, and dispatches the synchronous POST request.
- **Code:** Create `games/webhooks.py`:

```
import hmac
import hashlib
import json
import requests

def send_expiry_webhook(publisher, game_key_str, game_title, expired_at):
    # Define the JSON webhook payload data structure
    payload = {
        "event": "game_key.expired",
        "game_key": game_key_str,
        "game_title": game_title,
        "expired_at": expired_at.isoformat(),
    }

    # Secrets must be in bytes for cryptographic operations
```

```

secret = publisher.webhook_secret.encode('utf-8')

# Use sort_keys=True to guarantee consistent, deterministic string ordering before encoding
body = json.dumps(payload, sort_keys=True).encode('utf-8')

# Compute the HMAC SHA-256 signature using the secret key and encoded body bytes
signature = hmac.new(secret, body, hashlib.sha256).hexdigest()

headers = {
    "Content-Type": "application/json",
    # Standard header format: send signature with sha256= prefix
    "X-Signature": f"sha256={signature}",
}

try:
    # Dispatch synchronous POST request. timeout=5 prevents hanging indefinitely
    response = requests.post(publisher.webhook_url, json=payload, headers=headers, timeout=5)

    # Raise an exception if the response contains a 4xx or 5xx error status code
    response.raise_for_status()
except Exception as e:
    print(f"[Webhook] Delivery failed for publisher {publisher.id}: {e}")

```

- **Verify:** Check that the module imports compile without syntax errors.
- **⚠ Common Pitfalls:**
  - **Type errors with secrets:** Note that `hmac.new()` requires bytes, not strings. Both the `secret` and the `body` must be `.encode()`ed to bytes before hashing.

### Step 3.2: Updating the Management Command

- **Concept:** Modify `check_expired_keys.py` to trigger the synchronous webhook sender for every key we mark as expired.
- **Code:** Update `games/management/commands/check_expired_keys.py`:

```

from django.core.management.base import BaseCommand
from django.utils import timezone
from games.models import GameKey
from games.webhooks import send_expiry_webhook # Import webhook sender helper

class Command(BaseCommand):
    help = 'Mark expired keys and notify publishers synchronously.'

    def handle(self, *args, **options):
        # select_related performs SQL JOINS to fetch publisher details in a single query
        expired_keys = GameKey.objects.select_related('game__publisher').filter(
            expires_at__lte=timezone.now(),
            status='active'
        )

        # Bulk update status
        count = expired_keys.update(status='expired')

```

```
# Loop over expired keys and send webhook to publisher
for key in GameKey.objects.filter(status='expired').select_related('game__publisher'):
    send_expiry_webhook(key.game.publisher, key.key_string, key.game.title, key.expire

self.stdout.write(f'Expired {count} keys (sync webhooks sent).')
```

- **Verify:**

- Open `https://webhook.site` in a browser and copy the unique URL.
- Create a publisher in your Django admin panel and paste this URL into their `webhook_url` field.
- Expire a key in admin (by setting `expires_at` in the past and status to `active`).
- Run `python manage.py check_expired_keys`.
- Check the `webhook.site` logs to confirm that the HTTP POST request arrived with the correct JSON body and the `X-Signature` header.

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## 4. Discussion & Limitations (10 min)

Analyze the failures of synchronous webhooks:

1. **Thread Blocking:** The entire script blocks during the HTTP call.
  2. **Cascading Slowness:** If one publisher's server is slow, the script takes longer, delaying webhooks for all subsequent publishers.
  3. **No Resiliency (Retries):** If a publisher's server is down (connection error), the webhook is lost forever.
- *Preview: Tomorrow we will solve these three issues using an asynchronous task queue (Celery + Redis).*

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## 5. Check for Understanding (10 min)

**Question:** "A publisher claims they received a webhook from us but the payload was tampered with. How does HMAC protect against this?"

**Answer:** HMAC uses a shared secret to generate a cryptographic signature of the request body. If any character in the payload is altered during transit, the receiver's computed signature will not match the `X-Signature` header, revealing that the payload was modified and should be rejected.

**Question:** "Why do we pass `sort_keys=True` to `json.dumps()`? What could go wrong without it?"

**Answer:** JSON keys are unordered. If we don't sort keys, the serialization format might change (e.g. dictionary keys serialized in a different order), resulting in a different byte sequence. This would produce a different HMAC signature, causing verification to fail even if the content remains identical.

**Question:** "What happens to the management command if the publisher's server is down and we're making synchronous HTTP calls?"

**Answer:** The management command thread will block, waiting for the HTTP request to timeout (e.g. 5 seconds per request). If many keys expired for that publisher, or multiple publishers are down, the script will take a very long time to complete and could cause other pending updates to stall.

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## 6. Daily Checkpoint & Wrap-up (5 min)

- **Verification Criteria:**

- ☐ The `send_expiry_webhook` function is implemented in `games/webhooks.py`.
- ☐ Running `python manage.py check_expired_keys` dispatches an HTTP POST request to the publisher's webhook URL.
- ☐ The HTTP request header contains a valid `x-Signature` starting with `sha256=`.
- ☐ The webhook payload arrives at `webhook.site` successfully.